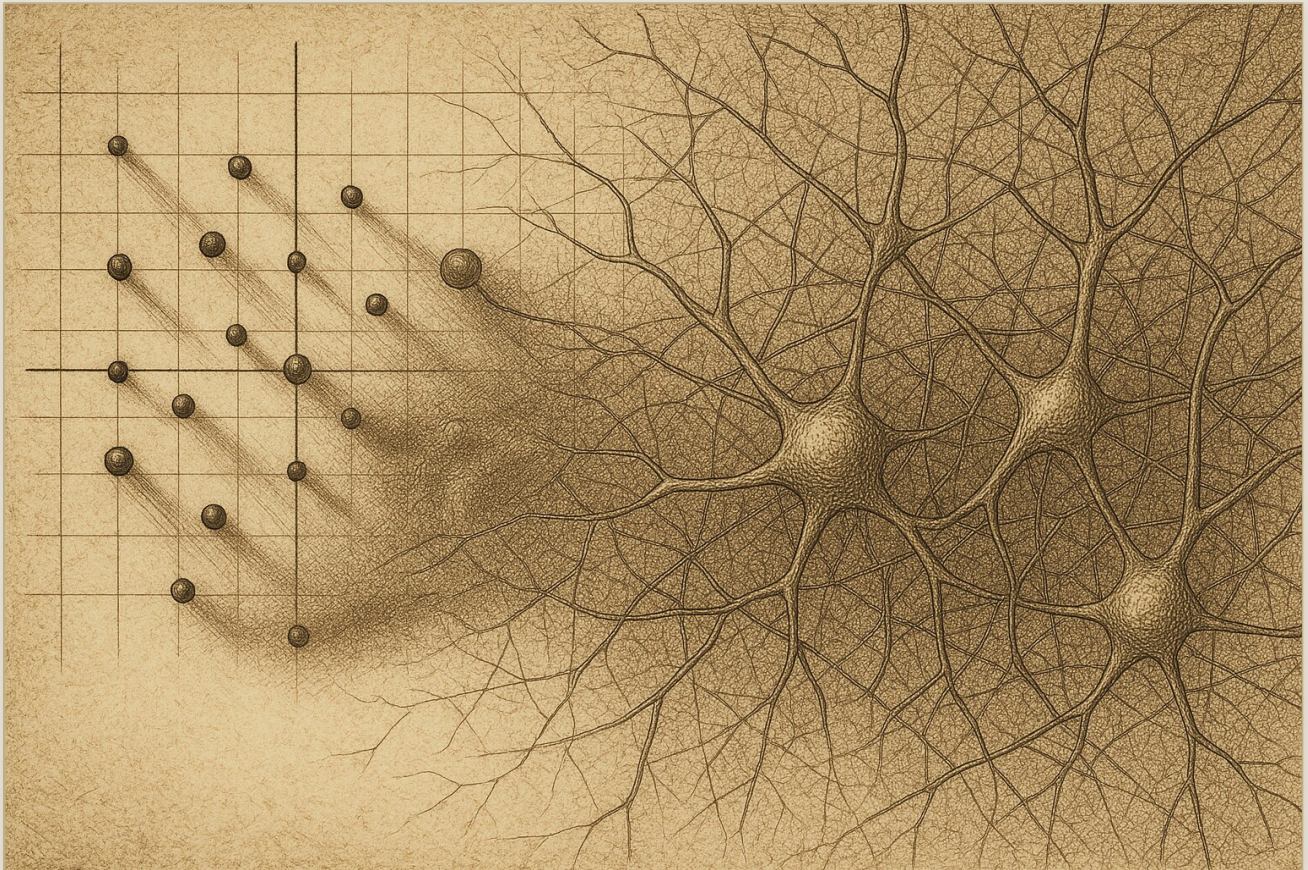




A NOCTURNAL DUEL · SCIENCE EXPLAINER

The Shadow of a *Spectrum*

Could you predict how long a sparse neural network holds an echo of its past just by reading off the eigenvalues of its wiring? For the long-run fate — yes. For the violent transients in between — the eigenvalues lie, and the truth hides in the operator's shadow.



On the left, the spectrum: eigenvalues plotted on the complex plane, each trailing the faint shadow of its pseudospectrum. On the right, the sparse non-normal network whose violent transients those shadows — not the eigenvalues alone — govern.

Date

10 June 2026

Topic

Whether, in non-Hermitian (non-normal) synaptic matrices, transient dynamics and effective memory are governed by the operator's non-normality (pseudospectrum / Kreiss constant / numerical radius) rather than by its eigenvalue spectrum

Claim type

Numeric — backed only by the attached reproducible compute bundle

Sides

Athos — defending the thesis · Aramis — refuting it · (public AI-generated sides; engine identities withheld)

Format

twelve alternating moves, every blow a simulation; an independent AI arbiter; an order-swap audit

Status: public adversarial AI stress-test. This is not peer review, not a scientific proof, and not an endorsement of the claim. Two AI-generated sides argued a frozen claim under a fixed protocol; an independent AI arbiter issued a structured verdict. For numerical claims, the reported numbers come only from the attached reproducible compute bundle. Publication means the duel passed the release gate; it does not promote the claim to discovery status. Passing the release gate means automated structural, integrity and format checks passed — it is not human methodological review, peer review, or approval of the claim's substance.

Epistemic status: scout / protocol pilot — exploratory. This issue is a scouting duel, **not an established scientific result**. Its decisive test runs on synthetic, controlled-spectrum operators: it probes whether the proposed mechanism is *sufficient in principle*, not whether real sparse synaptic matrices live in that regime. Treat the finding as a lead to investigate, not a confirmed claim about the seed paper's empirics.



network's wiring is a matrix, and the first thing anyone does with a matrix is find its *eigenvalues* — the numbers telling you how fast each natural mode grows or fades. For textbook systems that is the whole story: eigenvalues inside the unit circle mean activity dies away, outside it explodes. A recent paper on sparse synaptic matrices reads the network's behaviour — stability, transients, memory — directly off this eigenvalue distribution, treating the spectrum as the explanatory object.

The toolkit in one breath. A matrix is *normal* if its natural modes are mutually perpendicular; then eigenvalues govern everything. Real brains are not so polite: sparse, directed wiring makes the connectivity matrix *non-normal* — its modes overlap and lend each other energy. For such operators different numbers matter: the *pseudospectrum*, the *Kreiss constant* (a pseudospectral summary tied to transient growth), the *numerical radius*. The seed paper characterises the eigenvalues; this duel asks whether the eigenvalues are even the right object.

01 THE FROZEN CLAIM

The thesis Athos was assigned to defend, fixed before the duel and not edited afterwards:

"In sparse non-Hermitian synaptic matrices, transient dynamics and effective memory capacity are governed primarily by the non-normality of the operator (pseudospectrum / Kreiss constant / numerical radius), not by the eigenvalue distribution: two ensembles with an identical eigenvalue spectrum but different sparsity-induced non-normality will show systematically different transient amplification and memory."

The falsifier both sides accepted: construct ensembles matched on the eigenvalue spectrum (same circular-law radius, same spectral abscissa) but differing in non-normality, and measure transient amplification ($\sup_t \|A^t\|$) and memory capacity. If those metrics track the spectrum and ignore non-normality, the thesis is wrong.



THESIS · ATHOS

Eigenvalues set only the **asymptotic** fate. The violent transients and the memory live in the **pseudospectrum**. Hold the spectrum fixed and crank up non-normality,



COUNTER · ARAMIS

A single scalar of non-normality is too crude. When the spectrum is *truly* pinned, the supposed effect can vanish or run backwards; and memory depends on

and the transient amplification **climbs** — the eigenvalues never moved. The right yardstick is the Kreiss constant, not the spectral radius.

structure, noise and readout — not on how non-normal the matrix is. Show the eigenvalues are useless before discarding them.

02 THE DUEL: ARITHMETIC, NOT RHETORIC

Every move was a small matrix world with a **known** answer, so the two sides could compute who was right rather than argue it. The reproduction below rebuilds the cleanest version of each test. One half of the thesis was settled; the other was not.

The trap both sides fell into first

The obvious construction — perturb a Schur form — kept *moving the eigenvalues*, because a perturbation touching the diagonal blocks changes the very spectrum it should fix (our own first reproduction drifted the spectral radius from 0.85 to 1.21). The honest fix is a *strictly block-upper* perturbation that leaves every eigenvalue untouched. Only then is the test fair.

Settled: at a fixed spectrum, transients are governed by non-normality

With the spectrum pinned exactly — spectral radius **0.850 at every step** — we raise the non-normality (Henrici *departure*) from 0 to 27 and watch the worst-case transient amplification $\sup_t \|A^t\|$. It climbs from 0.85 to nearly a million. The eigenvalues never moved; the transients exploded.



Fig. 1 — Identical spectrum, exploding transients. The eigenvalue spectral radius is pinned at 0.85 (dashed blue, flat) while the worst-case transient amplification (oxblood, log scale) grows from 0.85 to $\sim 9.9 \times 10^5$ as non-normality rises. The eigenvalues are blind to this.
Source: 2026-06-10_cc_data.json (Experiment A).

The decisive measure: a pseudospectral sufficient statistic

Aramis's strongest move was fair: the crude scalar (Henrici departure) is *not* the right yardstick. Athos conceded it and named the correct object — the *Kreiss constant*, which the Kreiss matrix theorem binds two-sidedly to transient growth ($K \leq \sup_t \|A^t\| \leq e \cdot N \cdot K$). Pooling operators where *both* spectrum and non-normality vary, which sin-

gle number predicts the (log) transient? The pseudospectral measure wins decisively; the spectral radius is almost useless.

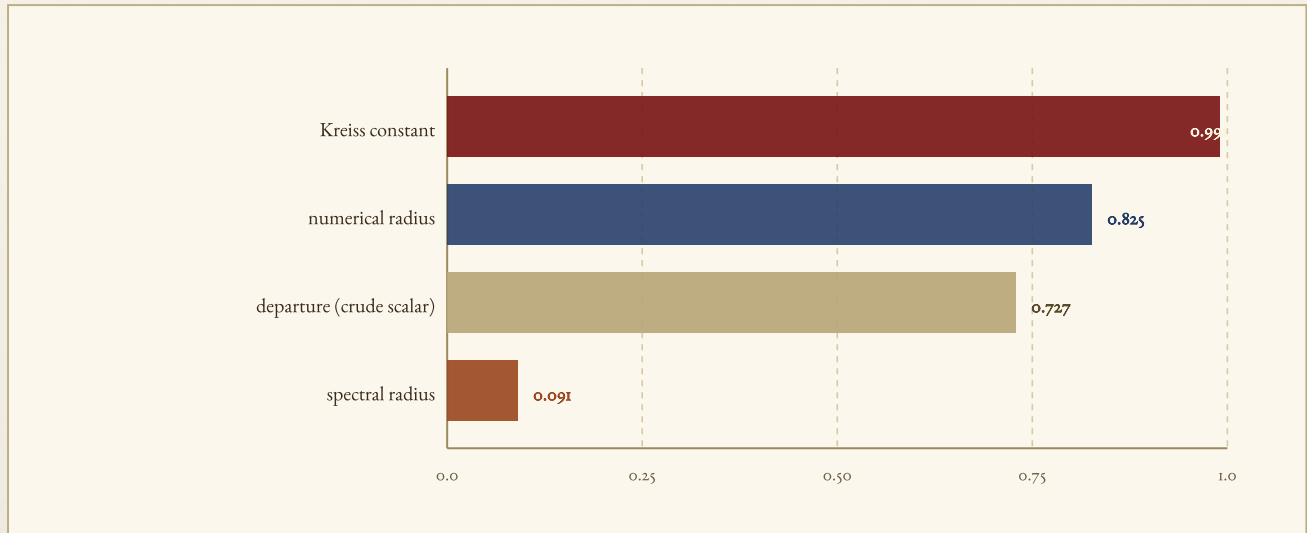


Fig. 2 — Which number predicts the transient? R^2 of log transient amplification regressed on each candidate, over 16 operators spanning four spectra and four non-normality levels. The Kreiss constant (pseudospectral) explains 99% of the variation; the eigenvalue spectral radius explains 9%. Source: `2026-06-10_cc_data.json` (Experiment B).

Where the thesis overreached: memory

Memory was the stubborn half. We compare two operators with an *identical* spectrum (radius 0.9): a **normal** one (departure 0) and a non-normal **delay-line chain** (departure 5.9). If non-normality governed memory as it governs transients, the chain should store far more. It does not — in this noisy linear readout the chain holds *less* (memory capacity **26.3** vs the normal operator's **33.8**). The lesson cuts both ways: the spectrum alone does not fix memory (the two differ at identical eigenvalues), but neither does a single non-normality scalar — memory also depends on structure, noise direction and readout.

03 THE VERDICT

OUTCOME (WHO ARGUED BETTER): PRO — THE THESIS SIDE PREVAILED ON ARGUMENTATION.

SUBSTANTIVE VERDICT (WAS THE CLAIM TRUE): UNCLEAR — TRUE FOR TRANSIENTS, OVERREACHING FOR MEMORY.

These two axes can diverge. A side won the debate; the claim's truth is split. Read "Outcome: PRO" as "the thesis was argued better here," not as "the whole claim is true."

The arbiter judged that the public side defending the thesis (Athos) argued better: it held the fixed-spectrum frame, repaired its own broken constructions, and shifted the contest from a crude scalar to the pseudospectrum and Kreiss constant — exactly the mathematically right object. The losing side (Aramis) landed real hits — it correctly flagged unpinned spectra and the crudeness of the departure scalar — but several of its decisive counter-examples were invalid: a "shuffle" that fed back the same matrix, and a dense construction that overflowed and changed the spectrum. **On the substance the verdict is split:** the transient half is well supported; the memory half is stronger than the data allow.

Basis: `real_compute` (attached bundle) · **Protocol:** PASS (12 moves, clean arbiter verdict) · **Order-swap audit:** PASS (substantive call stable on the natural run, "unclear" when the side order was swapped) · **Verbosity caveat:** the losing side wrote substantially more — length was

What this supports

- At an **identical** eigenvalue spectrum, transient amplification is governed by **non-normality**: pinning the spectral radius at 0.85, the worst-case transient grows from 0.85 to $\sim 10^6$ as departure rises.
- The right yardstick is **pseudospectral**: the Kreiss constant explains 99% of the (log) transient variation; the eigenvalue spectral radius explains 9%.
- The eigenvalue spectrum is **not a sufficient statistic** for transients — and becomes numerically unreliable precisely when non-normality is high.

What this does not support

- It does **not** show non-normality alone governs **memory**: at an identical spectrum the non-normal chain remembered *less*, not more.
- It does **not** claim the spectrum is useless — eigenvalues still set the **asymptotic** fate ($t \rightarrow \infty$).
- It does **not** establish that real sparse synaptic matrices live in this regime: the operators are controlled synthetic constructions, not biologically sampled connectomes.

04 WHERE THE LOSING SIDE STILL HAS A POINT

Aramis was right that a single non-normality scalar is too blunt — the memory experiment proves it: at a fixed spectrum, *structure* matters more than the raw magnitude of non-normality, and more non-normality did not buy more memory. Aramis also rightly insisted the clean claim only holds once the spectrum is *exactly* pinned. The honest next experiment — which neither side ran — is to take real sparse synaptic ensembles, control the spectrum strictly, and ask whether pseudospectral measures still dominate memory there.

05 PRIOR-WORK NOTE

This release makes **no originality claim**. The seed paper characterises the eigenvalue spectrum of sparse synaptic matrices and links sparsity type to stability and transients *through* the spectrum; the duel's contribution is a stress-test of the rival idea that non-normality, not the spectrum, is the governing object for transients and memory. The pseudospectral and memory results restate classical non-normal-dynamics theory (Trefethen & Embree; Ganguli, Huh & Sompolinsky) rather than report a new empirical finding. Novelty status: *not an originality claim*.



APPENDIX • REPRODUCIBILITY

Numbers and figures come from the attached repro bundle; full hashes, environment and rerun in **2026-06-10_compute_manifest.json**.

Construction: $A = Q(D + sT)Q^T$ with real orthogonal Q , a block-diagonal real Schur D fixing the eigenvalues, and a strictly *block*-upper-triangular T that raises non-normality while leaving the spectrum exactly fixed. $N = 100$ (A, C), $N = 80$ pooled (B). Diagnostics: Henrici departure; $\sup_t \|A^t\|_2$ ($t = 1..400$); numerical radius; discrete-time Kreiss constant on a finite grid outside the unit circle; memory capacity via a noisy linear reservoir (Ridge readout). Python 3.12.3 · numpy 2.4.4 · linux-x86_64 · seed 17.

Headline values: Experiment A — spectral radius fixed 0.850 across departures $0 \rightarrow 27.0$, $\sup \|A^t\|$ 0.85 $\rightarrow 9.95 \times 10^5$. Experiment B (16 operators) — R^2 of log sup on log Kreiss = 0.991, on numerical radius = 0.825, on departure = 0.727, on spectral radius = 0.091. Experiment C (identical spectrum 0.9) — memory capacity: normal 33.82, non-normal chain 26.27.

script_sha256: sha256:4f231c9e0a5ad81a16f1db9630d81c366ceea18b761bbff0f143e64ccd51f255 · **cc_data** **sha256:** sha256:5635fe4b700a7aoccbd9183b36d148161b624fdde5d726a88861f63689f749da

Rerun: python3 2026-06-10_repro.py $\rightarrow$ writes 2026-06-10_cc_data.json · **Errata / quarantine:** none.

REFERENCES

Ansari, M. G., & Shukla, P. (2026). On the synaptic matrix eigenvalues of sparsely connected neural networks (arXiv:2606.00326). arXiv. <https://arxiv.org/abs/2606.00326>

Seed publication — the eigenvalue characterisation of sparse synaptic matrices the duel was built on.

Ganguli, S., Huh, D., & Sompolinsky, H. (2008). Memory traces in dynamical systems. *Proceedings of the National Academy of Sciences*, 105(48), 18970–18975. <https://doi.org/10.1073/pnas.0804451105>

Supporting — memory and non-normal amplification in linear neural dynamics; the anchor for the memory half of the thesis.

Trefethen, L. N., & Embree, M. (2005). *Spectra and pseudospectra: The behavior of nonnormal matrices and operators*. Princeton University Press.

Supporting — pseudospectra and the Kreiss matrix theorem; the formal basis for using the Kreiss constant as the transient yardstick.

Citation style: APA 7. References ordered alphabetically by first author.

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IMPRIMATUR

O. D. · June 12, 2026

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